# RUBIN SCOUT

## Sprint Report 2

*Public Engagement Layer - Cosmic Animations*


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Sprint Duration: May 21, 2026 | Version: 1.0

github.com/Namrata-Modha/rubin-scout

## 1. Sprint Overview

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| **Sprint Name** | Public Engagement Layer - Cosmic Animations |
| **Sprint Number** | Sprint 2 |
| **Sprint Date** | May 21, 2026 |
| **Developer** | Namrata Modha |
| **Project** | Rubin Scout v0.1 |
| **Sprint Goal** | Make Rubin Scout accessible to anyone with no astronomy background by adding visual context to every alert type on the alert detail page |
| **Estimated Effort** | 6 hours |
| **Status** | Complete |
| **Live URL** | rubin-scout.vercel.app |
| **GitHub** | github.com/Namrata-Modha/rubin-scout |

## 2. Background and Problem Statement

### 2.1 The Accessibility Gap

Rubin Scout was built as a research tool and successfully reached its initial institutional audience - emails were sent to five research institutions including ARIES (India), Dunlap Institute (University of Toronto), McGill Space Institute, DRAO, and DAO Victoria. The CHIME/FRB project manager was engaged within hours of the first outreach.

However a significant secondary audience exists: curious members of the public who have no astronomy background but are drawn to the idea of real-time cosmic events. For this audience, the current alert detail page presents raw scientific data - ZTF identifiers, MJD timestamps, classification probabilities, and SIMBAD catalog names - without any visual context for what the object actually is or why it matters.

The inspiration for this sprint came from cinematic space visualizations on social media that sparked genuine curiosity about the universe in people with no prior astronomy interest. The hypothesis is that connecting real transient detections to a visual representation of the physical phenomenon they represent will make Rubin Scout meaningful to a much wider audience.

### 2.2 Strategic Product Pivot

This sprint represents a deliberate expansion of Rubin Scout's product positioning. Sprint 1 positioned Rubin Scout as a pipeline integration tool for working researchers. Sprint 2 expands the audience to include anyone who has ever looked up at the sky and wondered what is out there.

Both audiences are served by the same platform. The research dashboard and API remain unchanged. The cosmic animation section is additive - it appears on the alert detail page below the existing plain English description and above the telescope image, giving curious visitors a reason to stay while not disrupting the researcher workflow.

## 3. User Story

> *As someone with no astronomy background, I want to see a visual representation of what the detected object actually is so that I understand why this cosmic event matters and feel connected to the universe beyond my own planet.*

## 4. What Was Built

### 4.1 CosmicAnimation Component

A single React component (CosmicAnimation.jsx, 535 lines) that receives a classification string as a prop and renders a looping canvas animation specific to that object type. The component uses pure HTML5 Canvas with requestAnimationFrame - no new dependencies were added to the project.

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| **Component** | frontend/src/components/CosmicAnimation.jsx |
| **Integration point** | frontend/src/pages/AlertDetail.jsx - below description, above telescope image |

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| **Canvas height** | 200px, full card width |
| **Animation loop** | requestAnimationFrame with elapsed time in seconds |
| **Cleanup** | cancelAnimationFrame + ResizeObserver.disconnect() on unmount |
| **Resize handling** | ResizeObserver keeps canvas pixel-width in sync with container |
| **Determinism** | hf() hash function for per-particle values - no Math.random() in draw path |
| **New dependencies** | None |

## 4.2 Animation Catalogue

13 distinct animations covering all classification types present in the Rubin Scout dashboard. Each animation has a unique color atmosphere background, particle system, and caption line.

| Classification | Animation Description | Atmosphere Color | Caption Text |
|---|---|---|---|
| **Supernova Ia** | White dwarf accretes from blue companion, builds to detonation, golden-orange shockwave expands | Deep amber/gold | Watch a white dwarf steal matter until it explodes |
| **Supernova II** | Massive blue-white star pulses, collapses inward, red-orange shell explodes outward | Deep red-orange | Watch what happens when a massive star runs out of fuel |
| **Supernova Ibc** | Faster stripped core collapse, narrow cyan polar jets instead of spherical shell | Deep navy blue | A stripped stellar core collapsing and exploding |
| **SLSN** | Fills entire canvas with blinding white light before expanding blue-white corona | Deep ice blue | An explosion 100 times brighter than a normal supernova |
| **TDE** | 80 particles stream along star-to-black-hole-to-tidal-tail path continuously | Deep orange-black | Watch a star get torn apart by a black hole |
| **Kilonova** | Two neutron stars spiral inward, merge, gold and platinum particle burst expands | Deep gold-blue mix | Watch two neutron stars spiral together and collide |
| **AGN / QSO** | 120-particle accretion disk with elliptical orbits, pulsing bipolar relativistic jets | Deep blue-purple | Matter spiraling into a supermassive black hole |
| **Nucleus** | Same as AGN - active galactic nucleus with accretion disk and jets | Deep blue-purple | The intensely bright core of an active galaxy |
| **Black Hole Event** | Same as AGN - accretion disk and jets around central mass | Dark red-black | Matter and energy swirling around a black hole |

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|---|---|---|---|
| **CV / Nova** | Binary orbit, accretion stream from red giant to white dwarf, nova flash every 5s | Deep warm red | Two stars orbit each other, one feeding off the other |
| **Blazar** | 55 radial electric-blue streaks pulsing outward from center toward viewer | Deep electric blue | A jet of energy aimed directly at Earth |
| **FRB** | Expanding cyan pulse rings over static star field at 1.5s cadence | Deep teal-black | A millisecond pulse crossing billions of light years |
| **New Transient** | Faint drifting particles with occasional flicker - signal in noise | Near-black blue | Something new. We don't know what it is yet. |

## 4.3 Technical Implementation Details

### Animation Architecture

- Simple animations (SNIa, SNII, SNIbc, SLSN) are pure functions: drawFn(ctx, W, t) where t is elapsed seconds. No closure state needed.
- Stateful animations (TDE, KN, AGN, CV/Nova, Blazar, FRB, Default) use factory functions that return a tick closure, pre-computing per-particle arrays once at creation time rather than on every frame.
- All particle position and visual properties derived from hf(seed) deterministic hash - eliminates Math.random() from the draw path for consistent performance.

### Colored Atmosphere Backgrounds

- Each animation replaces the plain black background with a radial gradient atmosphere centered on the canvas midpoint.
- Gradient radiates from a classification-specific deep color at center to pure black at the edges, giving each object type an immediately recognizable visual identity.
- Background drawn first on every frame before particle systems, ensuring particles always render against the correct atmosphere.

### Caption Text System

- getCaptionText(cls) function maps each classification to a single plain English sentence explaining what the viewer is watching.
- Rendered as dim italic text directly above the canvas, matching the existing italic style in the description box.
- No heading, no card, no border - one quiet line connecting the description to the animation.

### Bottom Fade Overlay

- CSS gradient overlay fades the bottom 72px of the canvas from transparent to rgba(0,0,0,0.85).

- Ensures the animation blends seamlessly into the page background regardless of exact background color value.

## 5. Acceptance Criteria

| AC ID | Acceptance Criterion | Status | Notes |
|---|---|---|---|
| **AC-CA-01** | CosmicAnimation component renders without errors for all 13 classification strings | **Pass** | |
| **AC-CA-02** | Animation loops continuously without stopping or freezing after extended viewing | **Pass** | |
| **AC-CA-03** | Each classification renders a visually distinct animation with unique color atmosphere | **Pass** | |
| **AC-CA-04** | Unknown or null classification falls back to the default drifting particle animation | **Pass** | |
| **AC-CA-05** | Canvas resizes correctly when browser window is resized | **Pass** | ResizeObserver |
| **AC-CA-06** | requestAnimationFrame is cancelled on component unmount - no memory leaks | **Pass** | Cleanup verified |
| **AC-CA-07** | Caption text is readable, dim, italic, and matches existing page typography | **Pass** | |
| **AC-CA-08** | Bottom fade gradient blends animation into page background without a visible hard line | **Pass** | |
| **AC-CA-09** | No new npm dependencies added to the project | **Pass** | Pure Canvas only |
| **AC-CA-10** | Animation section appears between description box and telescope image on alert detail page | **Pass** | |
| **AC-CA-11** | Performance: canvas draws at 60fps on a standard mobile device without dropped frames | **Pending** | Mobile test needed |
| **AC-CA-12** | Kilonova animation displays the caption 'Where gold and platinum come from' during burst phase | **Pass** | |

## 6. Estimated Effort

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| **Animation logic (13 types)** | 3 hours - particle systems, timing, phase logic per classification |
| **Colored atmosphere backgrounds** | 30 minutes - radial gradient system, color mapping per type |
| **Caption text system** | 30 minutes - getCaptionText() function, styling |

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| **Integration into AlertDetail** | 30 minutes - import, placement, prop passing |
| **Visual tuning and brightness** | 1 hour - opacity, particle size, glow radius adjustments |
| **Testing across classifications** | 30 minutes - verify all 13 types render correctly |
| **Total** | 6 hours (single developer, single session) |

# 7. Product Impact

## 7.1 Audience Expansion

Prior to this sprint, Rubin Scout's practical audience was limited to astronomers and researchers who understood what classification strings like SNIa, KN, and TDE represented. The cosmic animation layer removes that prerequisite. Anyone who opens an alert detail page now receives an immediate visual explanation of the physical phenomenon regardless of their background.

## 7.2 Retention and Engagement

The alert detail page previously showed: plain English description, telescope image, SIMBAD cross-match, light curve, visibility planner. A curious visitor with no astronomy background would read the description and then encounter a graph of MJD values with no emotional context. The animation gives that visitor something to watch and feel before they engage with the scientific data below it.

## 7.3 Alignment with Product Vision

Rubin Scout's tagline is 'The cosmos for curious humans.' This sprint is the first feature that directly delivers on that tagline rather than serving it as an afterthought. The research functionality remains intact and unchanged. The public engagement layer sits on top of it.

# 8. Deferred Items

- Mobile performance validation - AC-CA-11 is marked Pending pending explicit mobile device testing at 60fps
- Sound design - ambient audio layer per classification type (pulsing for FRB, rumble for KN merger) was considered but deferred for a future sprint
- Animation intensity slider - letting users toggle between subtle and dramatic animation modes
- Share card - generating a shareable image with the animation frame and alert data for social media

## 9. Sprint Context

Sprint 2 follows Sprint 1 (ILMT Follow-Up API Endpoint, May 19, 2026) which built a programmatic pipeline integration endpoint for the ARIES/ILMT research team. The two sprints serve different users and different goals but together define the dual nature of Rubin Scout as a platform: deep enough for researchers, accessible enough for anyone.

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| **Sprint 1 goal** | Researcher workflow integration - ILMT Follow-Up Planner and API endpoint |
| **Sprint 2 goal** | Public engagement - visual context for all alert classifications |
| **Combined effect** | Rubin Scout serves both working researchers and curious members of the public from the same platform |
| **Next sprint candidates** | Sound design layer, share card generation, mobile performance optimization, CHORD telescope preset (next-gen CHIME at DRAO) |